

Proctor Guide

Reference answers & scoring — Case Studies 1–4

Cedarline Health Group is a regional healthcare payer-provider — roughly 8,000 employees across four hospitals and thirty-one clinics, plus a claims-processing business that handles the money. Growth by acquisition (including a second data centre absorbed as-is) and a mid-flight, disorderly move to public cloud have left a fragmented security posture. After a security incident in March, the board asked what would have stopped it — and leadership still cannot answer. Budget and executive attention are approved, but explicitly temporary, and the security team is small.

Use this guide to grade each team's workbook. For every pain point you'll find the expected product/platform, a model Solution-Engineer response, how it helps, and the **level cues** (Associate / Professional / Expert). Score each Case Study out of 100 (25 per pain point); 400 total.

Score each pain point by level (0–25)

Match each answer to the highest level it fully meets, then award points within that band. The per-pain-point **level cues** below each answer tell you what Associate / Professional / Expert looks like for that specific concern.

LEVEL	WHAT THE ANSWER DEMONSTRATES	POINTS
Expert	Correct product and capability; reasoning specific to Cedarline's constraints; states the outcome / how it helps; ties into the wider HMF design.	21–25
Professional	Correct product; explains how it addresses the concern with sound Cisco Secure / HMF reasoning; some Cedarline context.	14–20
Associate	Plausible product; identifies the problem, but reasoning is generic, partial, or product-name-only.	6–13
Not yet	Blank, off-topic, or a wrong product with no reasoning.	0–5

Level bands. Case Study /100 — Associate 40–64 · Professional 65–84 · Expert 85–100 (Developing <40). Overall /400 — Associate 160–259 · Professional 260–339 · Expert 340–400 (Developing <160).

Grade reasoning, not product-name-matching. A right product with no "how" caps at Associate; Cedarline-specific reasoning + outcome is what earns Expert. The companion **Proctor Grading Guide** has score lines and feedback boxes for use during grading.

1

CASE STUDY 1 · REFERENCE ANSWERS

AI Security

Focus. Cedarline is standing up AI across regulated workflows and public cloud faster than security can see it. Help them prove compliance, vet what they adopt, and get visibility and guardrails over AI traffic.

Pain Point 1

25 pts

Under the EU AI Act, we are classified as a 'Provider' for our internal AI systems. We need to demonstrate robustness and cybersecurity compliance to satisfy Article 15. How does Cisco help us meet these specific requirements?

PRODUCT /
PLATFORM**AI Defense** · AI Model and Application ValidationMODEL SE
RESPONSE

Cisco AI Defense is specifically designed to support Provider obligations under the EU AI Act. We utilize algorithmic red teaming technology for AI Model & App Validation, which identifies safety and security vulnerabilities at scale. By documenting these validation results and maintaining continuous runtime protection, you create a defensible, auditable framework.

HOW IT HELPS

It automates the technical evidence gathering required for "Conformity Assessments." By mapping AI-specific threats to standardized frameworks (like MITRE ATLAS and OWASP), it provides the documentation and continuous monitoring necessary to prove to regulators that your systems are robust and compliant.

LEVEL CUES

Associate (6–13) names AI Defense, generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Pain Point 2

25 pts

We are integrating various open-source models and third-party agents into our internal workflows. How can we ensure these assets aren't introducing vulnerabilities into our environment before they even go live?

PRODUCT /
PLATFORM**AI Defense** · Supply Chain Risk ManagementMODEL SE
RESPONSE

Cisco AI Defense addresses this through proactive Supply Chain Risk Management. We scan your model files, repositories, and MCP servers to identify malicious or unsafe assets. By integrating this into your CI/CD pipeline, we can block compromised AI assets before they are deployed, ensuring that your operational environment remains secure from the start.

HOW IT HELPS

It enables "Shift-Left" security by creating an inventory of all AI models and agents. By identifying vulnerabilities before production, it prevents the introduction of malicious code or poisoned data into your ecosystem.

LEVEL CUES

Associate (6–13) names AI Defense, generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

We know AI is showing up across our cloud environments faster than security can track it. We're worried we have models, agents, datasets, and MCP-connected tools running without visibility or consistent controls.

PRODUCT / PLATFORM	Cisco AI Defense - AI Cloud Visibility · Ai Cloud Visibility
MODEL SE RESPONSE	That's exactly the visibility gap Cisco AI Defense is built to solve. With AI Cloud Visibility, Cisco AI Defense continuously discovers AI assets across distributed cloud environments, including models, agents, datasets, MCP servers, and agent-to-tool workflows. Instead of relying on teams to manually identify what exists, you get a centralized view of your AI attack surface so you can find unsanctioned assets, assess whether they are protected, and bring them under policy.
HOW IT HELPS	identifies rogue or unsanctioned AI assets Maps relationships across data, models, agents, and tools Gives security teams a single-pane-of-glass view of AI exposure Helps close gaps before unknown AI usage becomes a production risk
LEVEL CUES	Associate (6–13) names Cisco AI Defense - AI Cloud Visibility, generic. · Professional (14–20) + how it addresses the concern. · Expert (21–25) + Cedarline-specific reasoning & outcome, ties to HMF.

We're running AI workloads in cloud accounts, but we don't have a reliable way to see which models and AI assets are actually in use. We're also concerned about workloads reaching out to third-party AI models without security knowing about it. And even when we do know the traffic path, we need a practical way to inspect and enforce guardrails on prompt and response traffic in the cloud without changing every application.

PRODUCT / PLATFORM	AI Defense = AI visibility and guardrail integration Multicloud Defense = cloud account connection, traffic visibility, policy attachment, and egress enforcement · AI Runtime Protection
MODEL SE RESPONSE	That's exactly what the AI Defense-Multicloud Defense integration is designed to address. Cisco AI Defense uses the integration to discover AI assets in your cloud environments and identify external AI destinations your workloads are calling. Then Cisco Multicloud Defense provides the cloud account connectivity, traffic visibility, and egress gateway enforcement point where AI Guardrails Profiles can be applied. In practice, that means you get AI asset visibility plus runtime inspection of prompt and response traffic at the cloud egress layer. AI Defense gives you the AI-specific visibility and guardrail logic, while Multicloud Defense gives you the cloud-network integration and enforcement point.
HOW IT HELPS	Finds AI assets in cloud accounts Identifies third-party AI model usage Uses existing cloud-network control points for enforcement Avoids relying only on application-level changes
LEVEL CUES	Associate (6–13) names AI Defense, generic. · Professional (14–20) + how it addresses the concern. · Expert (21–25) + Cedarline-specific reasoning & outcome, ties to HMF.

2

CASE STUDY 2 · REFERENCE ANSWERS

Cloud Edge

Focus. Workloads sprawl across AWS, Azure and on-prem, each with its own policy model. Contain lateral movement, prove consistent controls to auditors, and give Cedarline one place to define and enforce policy.

Pain Point 1

25 pts

If a breach occurs in one of our public cloud instances, we are terrified that the attacker will move laterally into our core on-premises systems. How can we contain these threats in a multicloud environment?

PRODUCT /
PLATFORM**Multicloud Defense** · Multicloud Defense GatewaysMODEL SE
RESPONSE

Multicloud Defense leverages macro and micro-segmentation capabilities to enforce strict communication boundaries. By integrating with our Hybrid Mesh Firewall and Hypershield, we can isolate workloads and inspect traffic between cloud segments, effectively stopping lateral movement before it reaches your sensitive data.

HOW IT HELPS

It provides granular visibility and control over east-west traffic. By applying Zero Trust principles across the hybrid fabric, you transform your network into a series of secure, isolated zones that prevent an attacker from pivoting through your infrastructure.

LEVEL CUES

Associate (6–13) names Multicloud Defense, generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Pain Point 2

25 pts

Our auditors require proof that our security controls are applied consistently across all our cloud workloads. Gathering this data from different cloud providers is a manual, time-consuming nightmare.

PRODUCT /
PLATFORM**Multicloud Defense** · Multicloud Defense VisibilityMODEL SE
RESPONSE

Cisco Multicloud Defense provides unified logging and traceability. Because our security framework is integrated across the network fabric, you can generate consolidated reports that map directly to your compliance requirements (such as NIS2 or DORA) across all cloud environments from a single dashboard.

HOW IT HELPS

It automates the collection of evidence for security audits. By consolidating logs and policy enforcement data, you create a "single source of truth" that simplifies compliance reporting and demonstrates proactive governance to regulators.

LEVEL CUES

Associate (6–13) names Multicloud Defense, generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Pain Point 3

25 pts

We have workloads spread across AWS, Azure, and on-premises data centers. Managing security policies individually in each cloud environment is creating massive operational overhead and leading to inconsistent security posture.

PRODUCT / PLATFORM

Multicloud Defense · Unified Security across public clouds

MODEL SE RESPONSE

Cisco Multicloud Defense, integrated within our Hybrid Mesh Firewall architecture, provides a unified control plane. You can define security policies once via Security Cloud Control and enforce them consistently across your entire distributed environment, regardless of the underlying cloud provider.

HOW IT HELPS

It replaces siloed, manual configuration with centralized orchestration. By abstracting the security policy from the cloud-native tools, it ensures that your security posture is uniform and eliminates the "configuration drift" that occurs when managing multiple cloud consoles.

LEVEL CUES

Associate (6–13) names Multicloud Defense, generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Pain Point 4

25 pts

We have workloads on-prem, in cloud, and in mixed environments, and policy consistency is becoming difficult. Every environment feels different, and we're worried about drift.

PRODUCT / PLATFORM

Cisco Secure Workload Cisco Secure Firewall Cisco Secure Firewall Management Center (FMC) Cisco Security Cloud Control

· Secure Workload Integration with Secure Firewall Management Center

MODEL SE RESPONSE

That's a common hybrid environment challenge. Cisco Secure Workload helps create a more unified segmentation model by using application dependency awareness and AI/ML-driven policy discovery to define policy based on how workloads actually communicate. Cisco Secure Firewall then enforces broader segmentation boundaries at the network layer.

HOW IT HELPS

Reduces policy drift across hybrid and multicloud environments
Aligns policy to application behavior, not just static IP-based rules
Supports consistent macro and micro controls together

LEVEL CUES

Associate (6–13) names Cisco Secure Workload Cisco Secure Firew..., generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

3

CASE STUDY 3 · REFERENCE ANSWERS

DC Edge — Perimeter Firewall

Focus. A mixed-vendor perimeter, an inherited rulebase and mostly-encrypted traffic leave blind spots and slow response. Restore visibility, speed the response, and modernise detection with the team they have.

Pain Point 1

25 pts

When our security tools detect a compromised device, our manual response time is too slow. By the time we identify the user and manually isolate the port, the attacker has already moved through the network.

PRODUCT /
PLATFORM**ISE** · Security Group Tags (pxGrid integration)MODEL SE
RESPONSE

Cisco ISE integrates with your security ecosystem via pxGrid and Adaptive Network Control (ANC). When a threat is detected, ISE can automatically trigger a quarantine policy, isolating the compromised device in real-time without requiring manual intervention.

HOW IT HELPS

It automates the "Response" phase of the incident lifecycle. By bridging the gap between detection and enforcement, it stops threats at the speed of the network.

LEVEL CUES

Associate (6–13) names ISE, generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Pain Point 2

25 pts

Our firewall environment has become too complex to manage manually.

PRODUCT /
PLATFORM**Cisco Security Cloud Control, FMC, FTD** · Unified Policy AdministrationMODEL SE
RESPONSE

Cisco applies AI-driven analysis to firewall telemetry and policy data to surface optimization opportunities, reduce manual troubleshooting, and guide cleaner policy administration.

HOW IT HELPS

Reduces manual policy review; identifies redundant/stale rules; lowers risk from human error.

LEVEL CUES

Associate (6–13) names Cisco Security Cloud Control, generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Pain Point 3

25 pts

Most of our traffic is encrypted, so we've lost visibility into a huge part of our environment.

PRODUCT / PLATFORM

Cisco Secure Firewall (FTD), FMC, Security Cloud Control · EVE

MODEL SE RESPONSE

Encrypted Visibility Engine (EVE) uses AI/ML-based fingerprinting to analyze encrypted traffic patterns and identify suspicious behavior without decrypting the payload.

HOW IT HELPS

Restores visibility into encrypted blind spots; avoids the performance/privacy overhead of decrypt-everything strategies.

LEVEL CUES

Associate (6–13) names Cisco Secure Firewall (FTD), generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Pain Point 4

25 pts

We're worried traditional signature-based detection won't catch new or unknown attacks quickly enough.

PRODUCT / PLATFORM

Cisco Secure Firewall (FTD), FMC, Security Cloud Control · Snort ML

MODEL SE RESPONSE

That's exactly where Snort ML helps. It adds machine-learning-based detection inside the Snort 3 engine to identify exploit behavior associated with unknown/zero-day threats.

HOW IT HELPS

Addresses the gap between known and unknown threat detection; reduces dependence on signature timing.

LEVEL CUES

Associate (6–13) names Cisco Secure Firewall (FTD), generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

4

CASE STUDY 4 · REFERENCE ANSWERS

Macro & Micro Segmentation

Focus. A flat data centre and fragmented Kubernetes networking invite east-west movement. Build segmentation Cedarline can adopt safely — with runtime visibility and application-dependency awareness so nothing breaks.

Pain Point 1

25 pts

Network policy helps, but we also need to know what's happening at runtime inside the workload. We need better detection of suspicious behavior and policy violations.

PRODUCT /
PLATFORM · Tetragon Runtime SecurityMODEL SE
RESPONSE

That's where Tetragon comes in. Tetragon extends the value of Cilium by providing eBPF-based runtime security and threat detection. It monitors system-level activity and network behavior in real time, so you can detect suspicious actions, investigate runtime events, and enforce security policy closer to the workload. In other words, Cilium helps control connectivity and segmentation, while Tetragon helps you understand and secure runtime behavior.

HOW IT HELPS

Adds runtime visibility beyond basic network controls
Detects suspicious system and network behavior in real time
Strengthens Kubernetes security with deeper workload-level insight

LEVEL CUES

Associate (6–13) names Isovalent, generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Pain Point 2

25 pts

Our Kubernetes networking feels fragmented across clusters and environments. We're dealing with too many moving parts, inconsistent policies, and too much operational complexity.

PRODUCT /
PLATFORM · Cilium Networking PoliciesMODEL SE
RESPONSE

That's exactly where Isovalent Cilium helps. Cilium provides a modern, eBPF-based Kubernetes networking layer that gives you consistent connectivity, policy enforcement, and visibility across clusters and environments. Instead of relying on older, IP-centric approaches that struggle in dynamic Kubernetes environments, Cilium enables more scalable and identity-aware networking designed for cloud-native workloads.

HOW IT HELPS

Replaces brittle, IP-centric networking models with a Kubernetes-native approach
Improves consistency across dynamic cluster environments
Reduces operational complexity with a more unified networking and policy layer

LEVEL CUES

Associate (6–13) names —, generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Pain Point 3

25 pts

Right now we're stitching together separate tools for networking, security, and observability in Kubernetes, and it's creating too much overhead.

PRODUCT / PLATFORM

cilium and tetragon

MODEL SE RESPONSE

That's exactly the consolidation opportunity Isovalent addresses. With Cilium for networking and policy, and Tetragon for runtime security, you get a more unified platform for connectivity, segmentation, observability, and security. This helps reduce tool sprawl and gives platform and security teams a more integrated operating model.

HOW IT HELPS

Consolidates multiple functions into one platform approach
Reduces tool sprawl and operational friction
Improves context sharing between networking, security, and operations teams

LEVEL CUES

Associate (6–13) names cilium and tetragon, generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Pain Point 4

25 pts

We know we need microsegmentation, but we don't have enough visibility into east-west flows or application dependencies to create policy safely. We don't want to break the application.

PRODUCT / PLATFORM

Cisco Secure Workload Cisco Secure Firewall · Application dependency mapping

MODEL SE RESPONSE

That's one of the biggest reasons customers adopt Cisco Secure Workload. It gives you deep visibility into workload communication and application dependency mapping, so you can see how applications actually behave before enforcing policy. That visibility helps you build segmentation policy with much more confidence, reduce guesswork, and move toward microsegmentation in a controlled way.

HOW IT HELPS

Reveals actual application communication paths
Reduces the risk of segmentation causing outages
Enables more precise policy design and staged enforcement

LEVEL CUES

Associate (6–13) names Cisco Secure Workload Cisco Secure Firew..., generic. · **Professional (14–20)** + how it addresses the concern. · **Expert (21–25)** + Cedarline-specific reasoning & outcome, ties to HMF.

Turning answers into a level

Use the **Level cues** under each pain point (above) with the master rubric from **Grading at a glance** to place every answer, then total the four pain points for the Case Study score and level.

LEVEL	WHAT THE ANSWER DEMONSTRATES	POINTS
Expert	Correct product and capability; reasoning specific to Cedarline's constraints; states the outcome / how it helps; ties into the wider HMF design.	21–25
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Case Study /100: Associate 40–64 · Professional 65–84 · Expert 85–100 (Developing <40). **Overall /400:** Associate 160–259 · Professional 260–339 · Expert 340–400 (Developing <160).

For a printable grading worksheet with per-pain-point score lines and “did well / improve” feedback boxes, use the **HMF Proctor Grading Guide** in this package.